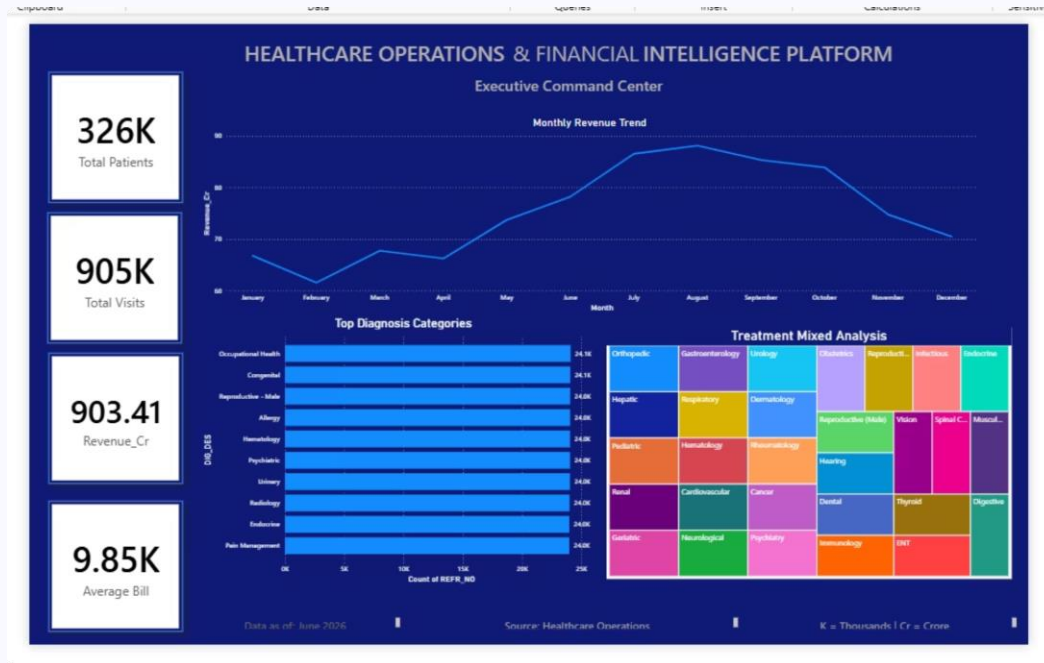




Healthcare Operations & Financial Analytics Platform

End-to-End Healthcare Analytics using **SQL**, **Python (Pandas)**, and **Power BI** – transforming raw operational and financial data into actionable business intelligence.

HEALTHCARE ANALYTICS

BUSINESS INTELLIGENCE

DATA ENGINEERING

The Business Challenge

Healthcare organizations generate vast volumes of patient, visit, diagnosis, treatment, and billing data – yet most lack the infrastructure to turn it into insight.

No Centralized Reporting

Data siloed across systems with no unified view

Limited Operational Visibility

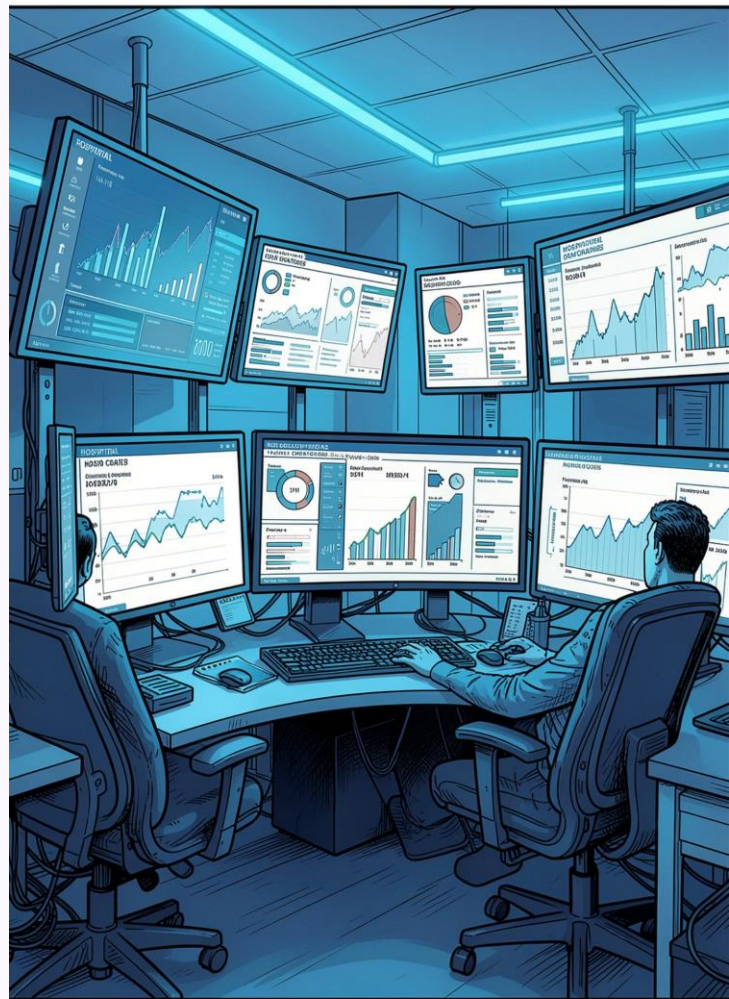
Poor tracking of patient utilization and throughput

Revenue Blind Spots

Difficulty monitoring billing trends and revenue performance

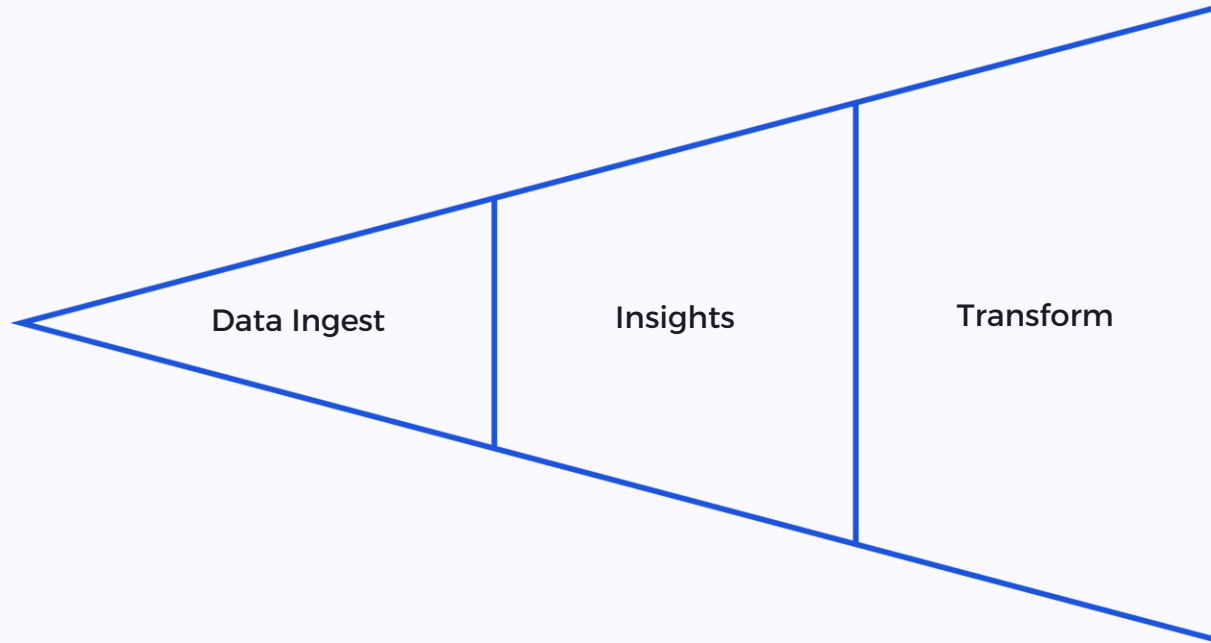
Diagnosis & Treatment Gaps

Limited insight into clinical patterns and treatment outcomes



End-to-End Analytics Pipeline

A structured, repeatable workflow transforms raw healthcare datasets into executive-level business intelligence – from ingestion to insight.



Each stage builds on the last: data is extracted from Excel, cleaned and queried in SQL, enriched with Python, visualized in Power BI, and delivered as actionable insights for healthcare leadership.

Technology Stack



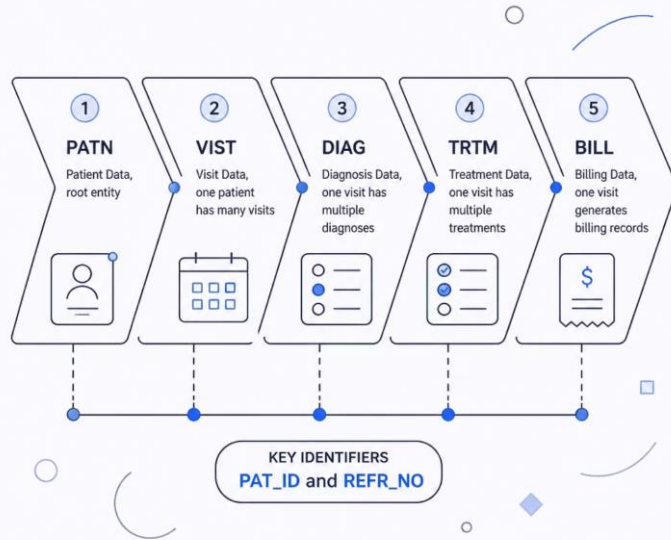
Why This Stack?

This combination represents the industry-standard toolkit for modern healthcare data analysts – covering the full spectrum from data extraction to executive reporting.

- **SQL** – complex querying and relational analysis
- **Python & Pandas** – programmatic cleaning and feature engineering
- **Power BI & DAX** – interactive dashboards and dynamic KPIs

Healthcare Data Model

Five integrated datasets linked through **PAT_ID** and **REFR_NO** form a relational model supporting end-to-end operational and financial analysis.



Dataset Overview

- **PATN** – Patient demographics and identifiers
- **VIST** – Visit records and encounter types
- **DIAG** – Diagnosis codes and categories
- **TRTM** – Treatment procedures and classifications
- **BILL** – Billing amounts and revenue records

i Key identifiers **PAT_ID** and **REFR_NO** enable seamless joins across all five datasets.

Data Quality & SQL Analysis

Data Quality Findings

- Missing values identified and systematically handled
- Mixed and invalid date formats detected and standardized
- Duplicate records identified and removed
- Relationship validation completed across all five datasets

SQL Concepts Applied

- Aggregations, Joins & Subqueries
- Common Table Expressions (CTEs)
- Window Functions & Ranking Functions
- Running Totals for cumulative revenue analysis

Business Impact: Improved data reliability for operational and financial reporting, enabling confident decision-making at the executive level.



Python & Pandas: Data Transformation



Techniques Applied

- Data cleaning, missing value handling & duplicate removal
- GroupBy analysis and pivot tables for segmentation
- Feature engineering and date analysis
- Exploratory Data Analysis (EDA) for pattern discovery

Key Functions Used

```
groupby() · agg() · merge() ·  
pivot_table() · apply() · lambda() ·  
fillna() · drop_duplicates() ·  
value_counts()
```

Power BI Executive Dashboard

A consolidated operational and financial command center delivering real-time visibility across the healthcare enterprise.

326K

Total Patients

Unique patients in the dataset

905K

Total Visits

Healthcare encounters recorded

₹903Cr

Total Revenue

Aggregate billing across all visits

₹9.8K

Average Bill

Per-visit billing benchmark

Dashboard components include monthly revenue trends, top diagnosis categories, treatment mix analysis, and revenue breakdowns by visit type and room – all designed for executive-level decision-making.

Analytics Insights at a Glance



Patient Analytics

Large patient population served with high utilization levels across visit types and demographics



Diagnosis Analytics

Diverse diagnosis categories identified with high-volume trends mapped for clinical planning



Treatment Analytics

Broad treatment distribution analyzed with treatment mix completed across all departments



Financial Analytics

Revenue performance tracked monthly with billing trends analyzed by visit type and room

Skills Demonstrated & Future Roadmap

Skills Demonstrated

- **Technical:** SQL, Python, Pandas, Power BI, DAX, Power Query
- **Analytical:** Data Modeling, EDA, Dashboard Development
- **Domain:** Healthcare Analytics, Financial Analytics, Business Intelligence

Potential Extensions

- Predictive Healthcare Analytics
- Revenue Forecasting Models
- Doctor Performance Analysis
- Patient Retention Analytics
- Department-Level Performance Monitoring

✔ **Project Outcome:** An end-to-end platform integrating SQL, Python, and Power BI to support operational efficiency, financial monitoring, and data-driven healthcare decision-making.